Design and Analysis of Aerodynamic and Thermal Solutions for Sidepods in a Student Formula SAE Vehicle

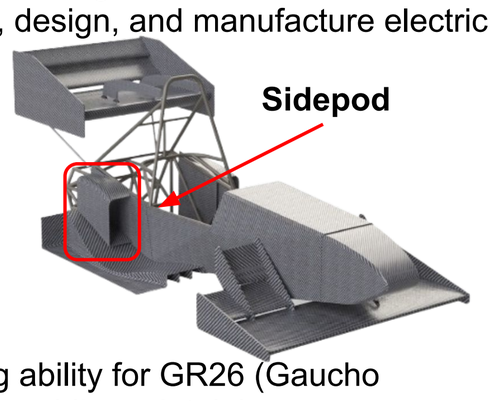
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Overview

Formula SAE is a collegiate design and engineering competition where students from across the continent research, design, and manufacture electric formula-style race cars.



In this project, we have taken the approach of performing **3D ANSYS simulations** on the sidepods, as well as developing designs that maximize the the mass flow rate of the sidepods.

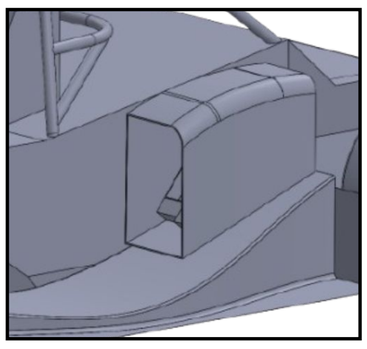
We use these findings to improve the cooling ability for GR26 (Gaucho Racing's 2026 car) and future vehicle designs while maintaining manufacturing feasibility and the Aero team's performance goals.

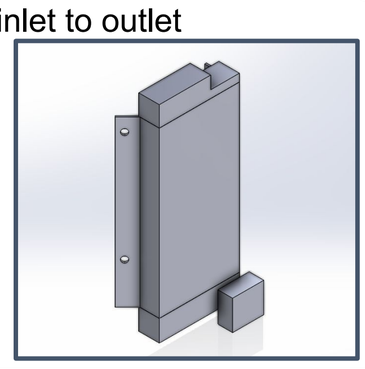
Aerodynamic Component

GR26 utilizes two aerodynamic sidepods, the primary focus of this research given limitations in the preliminary drivetrain cooling system.

Sidepod

- Large open inlet to allow maximum airflow through
- Smaller outlet to pull air through utilizing the pressure difference to feed more air through the radiator
- Smooth transition in internal ducting to prevent flow separation when going from inlet to outlet





Radiator

- Coolant flows through radiator tubes in a loop with the radiator, absorbing heat from the accumulator
- Air runs through the fins in the radiator to remove heat from the loop and cool the system

Design for Manufacturability (DFM)

To ensure the sidepod geometry can be produced with available resources, we outline the following composite manufacturing plan.

Mold Creation

- Laser-cut wooden profiles define cross-sections of the model.
- Foam is hotwire cut to shape, bonded, and sanded down to create a large male mold.



Layup Process

- Carbon fiber plies are trimmed to shape and laid onto the mold
- Epoxy resin is applied between layers until target thickness is achieved.
- The part is then placed in vacuum bagging to remove air and compact the laminate, reducing bubbles and creating a smoother finish
- The mold is given time to cure and harden before the composite is removed from the exterior.



Fit and Finish

- Excess material is trimmed.
- A clear coat layer is applied for protection.
- Holes are drilled for mounting tabs and quick release fasteners.
- Sidepods are integrated into cooling loop assembly.

Conclusion and Future Works

While the DFM study confirmed manufacturability and CFD analysis demonstrated a highly efficient design, the powertrain team's transition to a closed-loop cooling system eliminated the need for large radiator cores. Without a functional thermal requirement to justify their inclusion, the sidepods were rendered obsolete and removed from the current vehicle design. However, this project successfully established a robust, validated multiphysics simulation pipeline. We are enthusiastic about utilizing this foundational dataset to accelerate aerodynamic development and direct the sidepod integration for the upcoming GR27 vehicle.

Acknowledgements

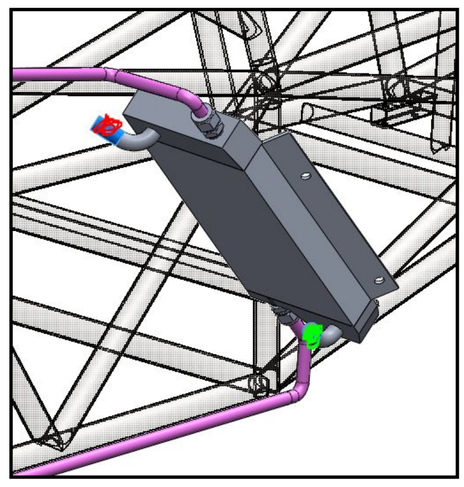
The Gaucho Racing Aerodynamics Team thanks URCA, our advisor Professor Emilie Dressaire, and our sponsors ANSYS and SolidWorks. Their support allowed for the success of this research project.

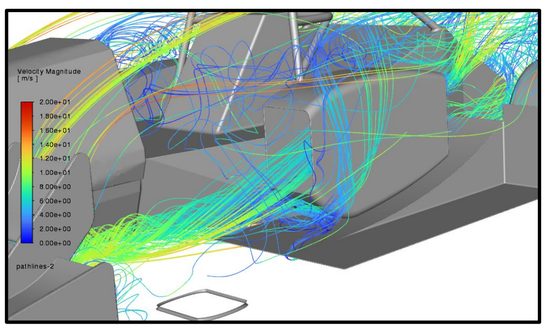
Computational Fluid Dynamics (CFD) For Downforce and Drag Analysis

To determine the most optimal shape for the sidepod, we performed a manual **Design of Experiments** on ANSYS Fluent, creating multiple sidepod iterations, each with a different radiator positioning and angling.

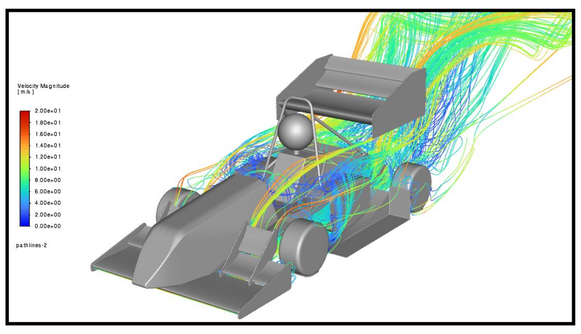
Model	Description
Case A	Angled 70° back from the horizontal lengthwise
Case B	Angled 70° back from the vertical lengthwise
Case C	Angled 70° forward from the horizontal widthwise
Case D	Angled 70° forward from the vertical widthwise

- We tested each case for coefficients of **lift**, coefficients of **drag**, and the **mass flow rate**.
- Our results yielded similar mass flow rates for each configuration, however the lift coefficients for **Cases A and C were much higher than Cases B and D**.
- Due to design constraints and cooling loop placement, we decided to move forward with **Case B**, which also had the lowest lift coefficient.





Sidepod Velocity Gradient



Full Car 3D Velocity Gradient

3D Simulations

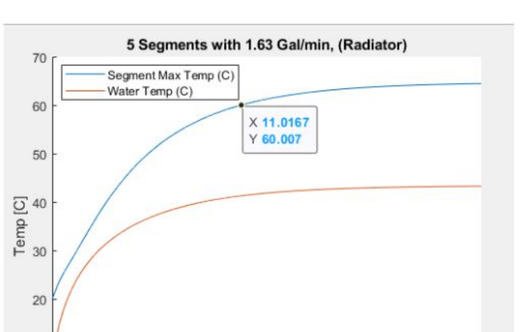
- 3D simulations are used to understand how the aerodynamics of the sidepod in conjunction with the diffusers affect the airflow around the entire car, as well as determining cooling performance of radiators.
- The optimized inlet geometry successfully captured **0.308 kg/s** of air. However, internal ducting expansion and boundary layer turbulence resulted in an effective mass flow rate of **0.216 kg/**s at the radiator face. This still represents a **46%** increase over the bare radiator baseline.
- Sidepods added negligible drag and contributed little to no loss of downforce in the rest of the aero components.

Trial	Mass Flow Rate
Radiator w/o Sidepod	**0.148 kg/s**
Radiator w/ Sidepod	**0.216 kg/s (+46% Increase)**

Comparisons of Radiator Performance

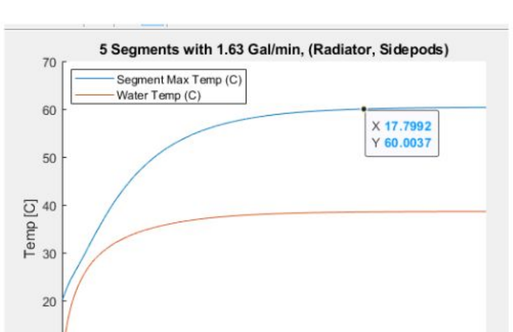
Sidepod Lift and Drag Penalty	Added Lift	Coefficient of Lift	Added Drag	Coefficient of Drag
Full Car Simulation w/ Sidepods	1.623 N	0.0196	0.322 N	0.00413

Water Cooling Loop Estimated Performance (SI)



Radiator:

- Air Speed: 6.47 m/s
- ~11.01 minutes @ max power



Radiator, Sidepods:

- Air Speed: 11.69 m/s
- ~17.91 minutes @ max power

According to the plots above, integrating sidepods increased system's airspeed by **81%**. This improved airflow enhances thermal management, increasing the car's endurance before overheating by **63%** and providing a much larger margin for endurance events.